

Building the Regression Model

Selection of Predictor Variables

Overview

- Data collection and preparation
- Reduction of explanatory or predictor variables
- Model refinement and selection
 - predictor variables must include *all* of the variables that actually predict the response
- Model validation
 - Stability and reasonableness of the regression model

All Possible Regressions Procedure

- Considers all possible subsets of the potential candidate predictor variables
- For example, if 10 predictor variables, $2^{10} = 1024$ possible regression models
- Select a subset of predictors that is 'best' according to some criteria
- use best subsets regression as a screening tool
- Further evaluate and refine the few models identified

Selection Criteria

- R^2
- adjusted R^2
- MSE
- Mallows' C_p -statistic
- PRESS statistic
- AIC, Akaike's Information Criterion
- AIC_c , Corrected AIC
- BIC, Bayesian Information Criterion

Selection Criteria - R^2

- $R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$
- R^2 always increases with the addition of predictor variables in the model
- The model with largest predictor variables will always have the highest R^2
- Look at the size of increase in R^2 with addition of predictor variables

$p \rightarrow$ number of predictors
in subset model
 $k \rightarrow$ total number of predictor variables

$n =$ total Sample Size

$p = 2$	$R^2 = .98$
$p = 3$	$R^2 = .985$
$p = 4$	$R^2 = .988$

Selection Criteria - R_a^2

- $R_a^2 = 1 - \frac{\frac{SSE}{n-p-1}}{\frac{SST}{n-1}} = 1 - \frac{n-1}{n-p-1} \frac{SSE}{SST}$
 $R_{adjusted}^2$
- p is the number of predictor in the current model
- choose the subset of the predictors that has the highest value of R_a^2
- choosing the subset of the predictors that has the highest value of R_a^2 *tends towards over-fitting*

Selection Criteria - MSE

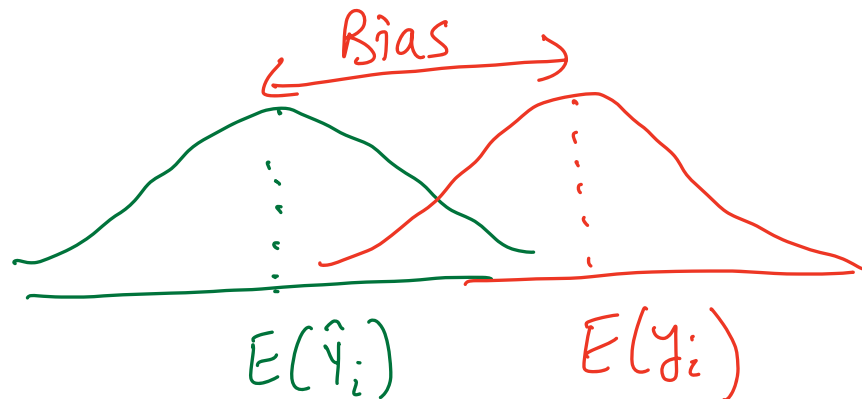
- $R_a^2 = 1 - \frac{\frac{SSE}{n-p-1}}{\frac{SST}{n-1}} = 1 - \frac{n-1}{n-p-1} \frac{SSE}{SST} = 1 - \frac{n-1}{SST} MSE$
- R_a^2 and MSE are equivalent methods
- As R_a^2 increases, MSE decreases
- choosing the subset of the predictors that has the smallest value of MSE

Selection Criteria -Mallows' C_p -statistic

- Two major issues in regression model
 - The **bias** in the predicted responses.
 - The **variation** in the predicted responses
- Estimates the size of the bias that is introduced into the predicted responses by having an underspecified model
- Bias in the predicted responses

$$B_i = E(\hat{y}_i) - E(y_i)$$

- Graph



Selection Criteria -Mallows' C_p -statistic

- Variation in the predicted responses
 - the random sampling variation - $\sigma_{\hat{y}_i}^2$
 - the variance associated with the bias $-B_i^2$
- To quantify the total variation in the predicted responses, we just sum the two variance components over n data points
- Standardize
- $\Gamma_p = \frac{1}{\sigma^2} \left\{ \sum_i^n \sigma_{\hat{y}_i}^2 + \sum_i^n ([E(\hat{y}_i) - E(y_i)]^2) \right\}$
- No bias in the predicted responses $\Gamma_p = p+1$

↓ ↓ intercept
Predictors

Selection Criteria -Mallows' C_p -statistic

- The best model is simply the model with the *smallest value of* Γ_p
- Estimate Γ_p

$$C_p = p + 1 + \left\{ \frac{MSE_p - MSE_F}{MSE_F} \right\} (n - p - 1)$$

$$C_p = \frac{SSE_p}{MSE_F} + 2(p + 1) - n$$

- For full model: $C_p = k + 1$

Selection Criteria -Mallows' C_p -statistic

- Subset models with small C_p values have a small estimated total (standardized) variation in predicted responses.
- When the C_p value is
 - near $p+1$, the bias is small (next to none)
 - much greater than $p+1$, the bias is substantial
 - below $p+1$, it is due to sampling error; interpret as no bias
- For the largest model containing all of the candidate predictors, $C_p = k+1$ (always). Not recommended

Selection Criteria -Mallows' C_p -statistic

Strategy

- Identify subsets of predictors for which the C_p value is **near $p+1$** (if possible).
- The full model always yields $C_p = k+1$, so don't select the full model based on C_p .
- If all models, except the full model, yield a large C_p not near $p+1$, it suggests some important predictor(s) are missing from the analysis. In this case, we are well-advised to identify the predictors that are missing!
- If a number of models have C_p near $p+1$, choose the model with the smallest C_p value, that will ensure that the combination of the bias and the variance is at a minimum.
- When more than one model has a small value of C_p value near $p+1$, in general, choose the simpler model or the model that meets your research needs.

Selection Criteria – PRESS statistic

Prediction Sum of Squares

- PRESS is a model validation method used to assess a model's predictive ability that can also be used to compare regression models
- PRESS is calculated by omitting each observation individually and then the remaining $n - 1$ observations are used to calculate a regression equation which is used to predict the value of the omitted response value
- Formula

$$PRESS = \sum_i^n (y_i - \widehat{y}_{i(i)})^2$$

- smaller the PRESS value, the better the model's predictive ability
- $R_{pred}^2 = 1 - \frac{PRESS}{SST}$
- A way to validate the predictive ability of your model without splitting the data into training and validation sets
- Both help prevent overfitting

Selection Criteria

Akaike's Information Criterion (AIC),

- $AIC_p = n [\ln(SSE)] - n [\ln(n)] + 2(p+1)$
- balances goodness of fit and a penalty for model complexity
- the smaller the value of AIC the better the model

Selection Criteria

Bayesian Information Criterion (BIC),

- $BIC_p = n [\ln(SSE)] - n [\ln(n)] + (p+1)[\ln(n)]$
- the penalty term in BIC is greater than the penalty term in AIC
- BIC penalizes complex models more heavily than AIC, thus favoring simpler models than AIC

Selection Criteria

AIC or BIC

R^2	R_a^2	MSE	C_p	PRESS	AIC	BIC
large	large	small	$p+1$	small	small	small

Hastie, Tibshirani and Freedman (2001)

- For model selection purposes, there is no clear choice between AIC and BIC
- BIC is asymptotically consistent as a selection criterion
 - given a family of models, including the true model, the probability that BIC will select the correct model approaches one as the sample size N approaches infinity
- AIC tends to choose models which are too complex as N approaches infinity
- For finite samples, BIC often chooses models that are too simple, because of the heavy penalty on complexity

Multiple Regression Model

Influential Diagnostics

Influence Diagnostics

- **Objective:** Identify those observations that exert disproportionate influence in statistical inference in MLR
- **Outliers:** data points which are not in sync with other data points
- Residuals $e_i = y_i - \hat{y}_i$

- Semi-Studentized Residuals

$$e_i^* = \frac{e_i}{\sqrt{MSE}}$$

HAT matrix

$$H = X (X^T X)^{-1} X^T \rightarrow \text{HAT Matrix}$$

$$\hat{Y} = H Y$$

$$e = (I - H) Y$$

h_{ii} is diagonal

h_{ij} is i^{th} row
and j^{th} column

$$\hat{V}(e_i) = \text{MSE} (1 - h_{ii})$$

$$\hat{\text{cov}}(e_i, e_j) = -h_{ij} \text{MSE}$$

Studentized Residuals

Appropriate to consider the magnitude of each e_i relative to its estimated standard deviation to give recognition to differences in the sampling errors of the residuals

$$r_i^{**} = \frac{e_i}{\sqrt{\text{MSE}(1-h_{ii})}}$$

While the residuals e_i will have substantially different sampling variations if their standard deviations differ markedly, the studentized residuals r_i have constant variance (when the model is appropriate).

Studentized Deleted Residuals

Residuals can be more effective for detecting outlying Y observations if we measure the i th residual e_i when the fitted regression is based on all of the cases except the i th one

$$d_i = y_i - \hat{y}_{i(-i)} \quad \text{delete } i^{\text{th}} \text{ observation}$$
$$d_i = \frac{e_i}{1 - h_{ii}}$$

Thus, deleted residuals will at times identify outlying Y observations when ordinary residuals would not identify these; at other times deleted residuals lead to the same identifications as ordinary residuals

$$\text{Var}(d_i) = \frac{\text{MSE}_{(-i)}}{1 - h_{ii}}$$

Studentized Deleted Residuals

Thus, the studentized deleted residuals t_i can be calculated from the residuals e_i , the error sum of squares SSE, and the hat matrix values h_{ii} , all for the fitted regression based on the n cases.

$$t_i = \frac{e_i}{\sqrt{MSE_{(-i)}(1-h_{ii})}} \sim t(n-p-1)$$

Test for Outliers

We identify as outlying Y observations those cases whose studentized deleted residuals are large in absolute value.

Bonferroni Test

Since we do not know in advance which case will have the largest absolute value $|t|$, we consider the family of tests to include n tests, one for each case. If the regression model is appropriate, so that no case is outlying because of a change in the model, then each studentized deleted residual will follow the t distribution with $n - P - 1$ degrees of freedom.

$$t_{(1-\frac{\alpha}{2n})} (n-p-1)$$

Identifying Outlying X observations

- The hat matrix, as we saw, plays an important role in determining the magnitude of a studentized deleted residual and therefore in identifying outlying Y observations.
- The hat matrix also is helpful in directly identifying outlying X observations. In particular, the diagonal elements of the hat matrix are a useful indicator in a multivariable setting of whether or not a case is outlying with respect to its X values.
- h_{ii} is a measure of the distance between the X values for the i th case and the means of the X values for all n cases.
- Thus, a large value h_{ii} indicates that the i th case is distant from the center of all X observations.
- The diagonal element h_{ii} in this context is called the leverage (in terms of the X values) of the i th case.

Identifying Outlying X observations

leverage value

$$h_{ii} > \frac{2(p+1)}{n}$$

$\Rightarrow$ outlying with respect
to X variable

Identifying influential cases

After identifying cases that are outlying with respect to their Y values and/or their X values, the next step is to ascertain whether or not these outlying cases are influential. We shall consider a case to be influential if its exclusion causes major changes in the fitted regression function.

Influence on single fitted value – DFFITS

- the DFFITS value for the i th case is a studentized deleted residual, increased or decreased by a factor that is a function of the leverage value for this case. If case i is an X outlier and has a high leverage value, this factor will be greater than 1 and (DFFITS) will tend to be large absolutely.
- As a guideline for identifying influential cases, consider a case to be influential if the absolute value of DFFITS exceeds 1 for small to medium data sets and $2^* \sqrt{\frac{p}{n}}$ for large data sets.

Identifying influential cases

Influence on ALL fitted value – Cook's Distance

Cook's distance measure considers the influence of the i th case on all n fitted values. Cook's distance measure, denoted by D_i , is an aggregate influence measure, showing the effect of the i th case on all n fitted values:

For interpreting Cook's distance measure, it has been found useful to relate D_i to the $F(p, n - p)$ distribution and ascertain the corresponding percentile value. If the percentile value is less than about 10 or 20 percent, the i th case has small influence on the fitted values. If, on the other hand, the percentile value is near 50 percent or more, the fitted values obtained with and without the i th case should be considered to differ substantially, implying that the i th case has a major influence on the fit of the regression function.

Identifying influential cases

Influence on Regression Coefficients – DFBETAS

A measure of the influence of the i th case on each regression coefficient b_k ($k = 0, 1, \dots, p - 1$) is the difference between the estimated regression coefficient b_k based on all n cases and the regression coefficient obtained when the i th case is omitted

- The DFBETAS value by its sign indicates whether inclusion of a case leads to an increase or a decrease in the estimated regression coefficient
- its absolute magnitude shows the size of the difference relative to the estimated standard deviation of the regression coefficient.
- A large absolute value of $(\text{DFBETAS})_{k(i)}$ is indicative of a large impact on the i th case on the k th regression coefficient.
- As a guideline for identifying influential cases, a case influential if the absolute value of DFBETAS exceeds 1 for small to medium data sets and $2/\sqrt{n}$ for large data sets.

Identifying influential cases

- It is usually a good idea to examine in a direct fashion the inferences from the fitted regression model that would be made with and without the case(s) of concern.
- If the inferences are not essentially changed, there is little need to think of remedial actions for the cases diagnosed as influential
- On the other hand, serious changes in the inferences drawn from the fitted model when a case is omitted will require consideration of remedial measures.